DIFFERENTIATION

Differentiation

- $f(x) = c$, $\frac{dy}{dx} = 0$
- $f(x) = x^n$, $\frac{dy}{dx} = nx^{n-1}$
- Chain rule

  $y = fg(x)$, $u = g(x)$, $y = f \times u$

  $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
- $y = cu$, c is constant, $\frac{dy}{dx} = c \times \frac{du}{dx}$
- $y = u + v$, $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$
- Product Rule

  $y = u \times v$, $\frac{dy}{dx} = uv' + vu'$
- Quotient Rule

  $y = \frac{u}{v}$, $\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$
- the deriavative of exponential function
  - $\frac{d}{dx}[e^u] = e^u \times u'$, $\ln(e) = 1$

    example: $\frac{d}{dx}[e^{6x^2}] = e^{6x^2} \times 12x = 12xe^{6x^2}$
  - $\frac{d}{dx}[a^u] = a^u \times u' \times \ln(a)$, "a" is a number

    example: $\frac{d}{dx}[3^{2x^4}] = 3^{2x^4} \times 8x^3 \times \ln 3$
- the deriavative of logarithms function
  - $\frac{d}{dx}[\ln u] = \frac{u'}{u}$, $\ln u^x = x \ln u$
  - $\frac{d}{dx}[\log_a u] = \frac{u'}{u \times \ln a}$
- implicit differentiation
  - need to differentiate both side
  - when diffentiate "y", need to add "$\frac{dy}{dx}$" after it
  - example: $x^2 + y^2 = 12x$

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(12x)$$

$$\frac{d}{dx}x^2 + \frac{d}{dx}y^2 = \frac{d}{dx}12x$$

$$2x + 2y\left(\frac{dy}{dx}\right) = 12$$

$$\frac{dy}{dx} = \frac{12 - 2x}{2y} = \frac{6 - x}{y}$$

$\therefore$ $\frac{d}{dx}\sin x = \cos x$

$\frac{d}{dx}\cos x = -\sin x$

$\frac{d}{dx}\tan x = \sec^2 x$

Natural logarithm rules and properties

Rule name	Rule
Product rule	$\ln(x \cdot y) = \ln(x) + \ln(y)$
Quotient rule	$\ln(x / y) = \ln(x) - \ln(y)$
Power rule	$\ln(x^y) = y \cdot \ln(x)$
ln derivative	$f(x) = \ln(x) \Rightarrow f'(x) = 1 / x$
ln integral	$\int \ln(x)dx = x \cdot (\ln(x) - 1) + C$

Original Rule	Generalized Rule (Chain Rule)
$\frac{d}{dx}\sin x = \cos x$	$\frac{d}{dx}\sin \mathbf{u} = \cos \mathbf{u} \frac{du}{dx}$
$\frac{d}{dx}\cos x = -\sin x$	$\frac{d}{dx}\cos \mathbf{u} = -\sin \mathbf{u} \frac{du}{dx}$
$\frac{d}{dx}\tan x = \sec^2 x$	$\frac{d}{dx}\tan \mathbf{u} = \sec^2 \mathbf{u} \frac{du}{dx}$
$\frac{d}{dx}\cot x = -\operatorname{cosec}^2 x$	$\frac{d}{dx}\cot \mathbf{u} = -\operatorname{cosec}^2 \mathbf{u} \frac{du}{dx}$
$\frac{d}{dx}\sec x = \sec x \tan x$	$\frac{d}{dx}\sec \mathbf{u} = \sec \mathbf{u} \tan \mathbf{u} \frac{du}{dx}$
$\frac{d}{dx}\operatorname{cosec} x = -\operatorname{cosec} x \cot$	$\frac{d}{dx}\operatorname{cosec} \mathbf{u} = -\operatorname{cosec} \mathbf{u} \cot \mathbf{u} \frac{du}{dx}$

The first principle of differentiation

$$f'(x) = \lim_{\delta x \to 0} \frac{f(x+\delta x) - f(x)}{\delta x},$$

- when the limit is exist.

- The procedure is known as the differentiation from the **first principle or called as definition of differentiation.**

The first principle

- Step 1: Given $y = f(x)$. Write an expression $f(x + \delta x)$.

- Step 2: Obtain the expression difference between $f(x + \delta x)$ and $f(x)$

$$f(x + \delta x) - f(x)$$

- Step 3: Simplify the expressions

$$\frac{f(x + \delta x) - f(x)}{\delta x}$$

- Step 4: Find the limit

$$f'(x) = \lim_{\delta x \to 0} \frac{f(x + \delta x) - f(x)}{\delta x}$$

Example

- By using differentiation from the first principle, find the derivatives of the following function

- a) $y = 2x^2$
- Given that $f(x) = 2x^2$
  Step 1: $f(x + \delta x) = 2(x + \delta x)^2 = 2x^2 + 4x\delta x + 2(\delta x)^2$
- Step 2: $f(x + \delta x) - f(x) = 2x^2 + 4x\delta x + 2(\delta x)^2 - 2x^2$
  $= 4x\delta x + 2(\delta x)^2$
- Step 3: $\frac{f(x+\delta x)-f(x)}{\delta x} = \frac{4x\delta x+2(\delta x)^2}{\delta x} = 4\mathrm{x} + 2\delta x$
- Step 4: $f'(x) = \lim_{\delta x \to 0} \frac{f(x+\delta x)-f(x)}{\delta x}$
  $= \lim_{\delta x \to 0} 4\mathrm{x} + 2\delta x = 4x$

Example

- b) $y = \dfrac{1}{x}$

- Given that $f(x) = \frac{1}{x}$

  Step 1: $f(x + \delta x) = \frac{1}{x+\delta x}$

- Step 2: $f(x + \delta x) - f(x) = \frac{1}{x+\delta x} - \frac{1}{x}$

$$= -\frac{\delta x}{x^2 + xdx}$$

- Step 3: $\frac{f(x+\delta x) - f(x)}{\delta x} = -\frac{\delta x}{x^2 + x\delta x} \cdot \frac{1}{\delta x}$

$$= -\frac{1}{x^2 + x\delta x}$$

- Step 4: $f'(x) = \lim\limits_{\delta x \to 0} \frac{f(x+\delta x) - f(x)}{\delta x}$

$$= \lim_{\delta x \to 0} -\frac{1}{x^2 + x\delta x} = -\frac{1}{x^2}$$

Example

- c) $y=\sqrt{x+5}$

- Given that $f\left(x\right)=\sqrt{x+5}$

$$f'(x)=\frac{f\left(x+\partial x\right)-f\left(x\right)}{\partial x}$$

$$=\frac{\sqrt{x+\partial x+5}-\sqrt{x+5}}{\partial x}$$

$$=\frac{\left(\sqrt{x+\partial x+5}-\sqrt{x+5}\right)}{\partial x}\cdot\frac{\left(\sqrt{x+\partial x+5}+\sqrt{x+5}\right)}{\left(\sqrt{x+\partial x+5}+\sqrt{x+5}\right)}$$

$$=\frac{x+\partial x+5-\left(x+5\right)}{\partial x\left(\sqrt{x+\partial x+5}-\sqrt{x+5}\right)}$$

$$=\frac{1}{2\sqrt{x+5}}$$

Practise 2.1

- By using differentiation from first principle, find the derivatives

  a) $y = x + x^2$

  b) $y = \sqrt{2x - 1}$

  c) $y = x - \frac{1}{x^2}$

  d) $y = \frac{1}{1+x}$

  e) $y = \frac{1-x}{2x}$

Answer

a) $1 + 2x$
b) $\frac{1}{\sqrt{2x-1}}$
c) $1 + \frac{2}{x^3}$
d) $-\frac{1}{(1+x)^2}$
e) $-\frac{1}{2x^2}$

Solution (b)

b) $\sqrt{2x-1}$

① $f(x+\partial x) = \sqrt{2(x+\partial x)-1}$

② $f(x+\partial x) - f(x) = \sqrt{2(x+\partial x)-1} - \sqrt{2x-1}$

$$= \sqrt{2(x+\partial x)-1} - \sqrt{2x-1} \cdot \frac{\sqrt{2(x+\partial x)-1} + \sqrt{2x-1}}{\sqrt{2(x+\partial x)-1} + \sqrt{2x-1}}$$

$$= \frac{2(x+\partial x)-1-(2x-1)}{\sqrt{2(x+\partial x)-1} + \sqrt{2x-1}} = \frac{2x+2\partial x-1-2x+1}{\sqrt{2(x+\partial x)-1} + \sqrt{2x-1}}$$

$$= \frac{2\partial x}{\sqrt{2(x+\partial x)-1} + \sqrt{2x-1}}$$

③ $$\frac{f(x+\partial x)-f(x)}{\partial x} = \frac{2\cancel{\partial x}}{\cancel{\partial x}\left(\sqrt{2(x+\partial x)-1} + \sqrt{2x-1}\right)}$$

$$= \frac{2}{\sqrt{2(x+\partial x)-1} + \sqrt{2x-1}}$$

④ $$\lim_{\partial x \to 0} \frac{f(x+\partial x)-f(x)}{\partial x} = \frac{2}{\sqrt{2(x+0)-1} + \sqrt{2x-1}}$$

$$= \frac{\cancel{2}}{\cancel{2}\sqrt{2x-1}} = \frac{1}{\sqrt{2x-1}}$$ #

Solution (c)

$$x - \frac{1}{x^2}$$

$$f(x+\partial x) = (x+\partial x) - \frac{1}{(x+\partial x)^2}$$

$$f(x+\partial x) - f(x) = (\cancel{x}+\partial x) - \frac{1}{(x+\partial x)^2} - \left(\cancel{x} - \frac{1}{x^2}\right)$$

$$= \partial x - \frac{1}{(x+\partial x)^2} + \frac{1}{x^2}$$

$$= \frac{\partial x(x^2)(x+\partial x)^2 - x^2 + (x+\partial x)^2}{x^2(x+\partial x)^2}$$

$$= \frac{\partial x(x^2)(x^2 + 2x\partial x + (\partial x)^2 - \cancel{x^2} + \cancel{x^2} + 2x\partial x + (\partial x)^2}{x^2(x+\partial x)^2}$$

$$= \frac{x^4\partial x + 2x^3(\partial x)^2 + (x^2)(\partial x)^3 + 2x\partial x + (\partial x)^2}{x^2(x+\partial x)^2}$$

$$\frac{f(x+\partial x) - f(x)}{\partial x} = \frac{\cancel{\partial x}(x^4 + 2x^3\partial x + x^2(\partial x)^2 + 2x + \partial x)}{\cancel{\partial x} \cdot x^2(x+\partial x)^2}$$

$$\lim_{\partial x \to 0} \frac{f(x+\partial x) - f(x)}{\partial x} = \frac{x^4 + 0 + 0 + 2x + 0}{x^4 + 0 + 0} = \frac{x^4 + 2x}{x^4}$$

$$= 1 + \frac{2}{x^3}$$ #

Differentiation of Simple Algebraic Function

Theorem 2.1 (The derivative of a constant Function)
if $y = c$ where c is a constant, then

$$\frac{dy}{dx} = 0$$

Theorem 2.2 (The Power rule for Positive Integers)
if $y = x^n$ and n is a positive integer, then

$$\frac{dy}{dx} = nx^{n-1}$$

The following is the table of derivatives of some basic functions

$f(x)$	$f'(x)$
c	0
$mx + c$	m
x^a	ax^{a-1}
e^x	e^x
$\ln x$	$\frac{1}{x}$

Example

1. $y = x^{15}$

$$\frac{dy}{dx} = 15 \cdot x^{15-1} = 15x^{14}$$

2. $y = 4x^3 + 2x^2 + 3x$

$$\frac{dy}{dx} = (3)(4)x^{3-1} + (2)(2)x^{2-1} + 3x^{1-1}$$
$$= 12x^2 + 4x + 3$$

3. $y = \frac{1}{x^5} - \frac{2}{x^3} + \frac{x^3}{2}$
$$= x^{-5} - 2x^{-3} + \frac{x^3}{2}$$

$$\frac{dy}{dx} = (-5)x^{-5-1} - (-3)(2)x^{-3-1} + (3)\frac{x^{3-1}}{2}$$
$$= \frac{-5}{x^6} + \frac{6}{x^4} + \frac{3x^2}{2}$$

4. $y = 3\sqrt{x} + \sqrt[3]{x}$
$$= 3(x)^{\frac{1}{2}} + (x)^{\frac{1}{3}}$$

$$\frac{dy}{dx} = \frac{3}{2}x^{-\frac{1}{2}} + \frac{1}{3}x^{-\frac{2}{3}}$$
$$= \frac{3}{2\sqrt{x}} + \frac{1}{3\sqrt[3]{x^2}}$$

Practise 2.2

Differentiate all the following:

a) $y = x^4 + \frac{4}{x^2}$

b) $y = \frac{1}{x} + 3x^2 - \sqrt[4]{x}$

c) $y = \sqrt{x-2} + 3\sqrt[3]{x}$

d) $y = \frac{x^2+2}{x^3}$

e) $y = x^2(1 + \frac{1}{x})$

Answer

a) $y = 4x^3 - \frac{8}{x^3}$

b) $y = -\frac{1}{x} + 6x - \frac{1}{4\sqrt[4]{x^3}}$

c) $y = \frac{1}{2\sqrt{x-2}} + \frac{1}{\sqrt[3]{x^2}}$

d) $y = -\frac{1}{x^2} - \frac{6}{x^4}$

e) $y = 2x + 1$

The Chain Rule

- The chain rule is used to calculate derivatives of composites functions such as $f\big(g(x)\big)$.

- If $y = f\big(g(x)\big)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Example

- Differentiate the following function with respect to x

a) $y = (x^3 + 6x - 3)^{20}$

Let $u = x^3 + 6x - 3$ and $y = u^{20}$. By using the chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= (20u^{19}) \cdot (3x^2 + 6)$$

$$= 20(x^3 + 6x - 3)^{19}(3x^2 + 6)$$

Example

b) $y = \sin(2x - 5)$

Let $u = 2x - 5$ and $y = \sin(u)$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= \cos(u) \cdot 2$$

$$= 2\cos(2x - 5)$$

c) $y = e^{3x+1}$

Let $u = 3x + 1$ and $y = e^u$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{d}{dx}[e^{3x+1}]\frac{d}{dx}[3x + 1]$$

$$= 3e^{3x+1}$$

d)) $y = \ln 3x$

Let $u = 3x$ and $y = \ln u$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{d}{dx}[\ln 3x]\frac{d}{dx}[3] = \frac{1}{3x} \cdot 3 = \frac{1}{x}$$

Practise 2.3

- a) $y = \sqrt{1 - 2x}$
- b) $y = \frac{(x+1)^2}{(x+2)^3}$
- c) $y = (2x - 1)^3(3x + 2)^4$
- d) $y = e^{5x + \cos x}$
- e) $y = 3\ln(x^2 + 2x)$

Answer

- a)$-\frac{1}{\sqrt{1-2x}}$
- b)$\frac{1-x^2}{(x+2)^4}$
- c)$42x(2x-1)^2(3x+2)^3$
- d) $(5 - sinx\ )e^{5x+\cos x}$
- e) $\frac{6(x+1)}{x(x+2)}$

Differentiation Rules

1. Differentiation of Multiples

- Let $y = cu$, u is a function of x and c is a constant,

$$\frac{dy}{dx} = c\frac{du}{dx}$$

2. Differentiation of Sums

- Let u and v be differentiable function of x. If $y = u + v$,

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

Example

a) $y = 4x^9$

$$\frac{dy}{dx} = 4\frac{d}{dx}[x^9] = 4(9x^8) = 36x^8$$

b) $y = 4x^3 + 2x^2 + 3x$

$$\frac{dy}{dx} = 4\frac{d}{dx}[x^3] + 2\frac{d}{dx}[x^2] + 3\frac{d}{dx}[x]$$

$$= 4(3x^2) + 2(2x) + 3 = 12x^2 + 4x + 3$$

c) $y = 2\tan x - 3\cos x$

$$\frac{dy}{dx} = 2\frac{d}{dx}[\tan x] - 3\frac{d}{dx}[\cos x]$$

$$= 2sec^2x - 3(-\sin x)$$

$$= 2sec^2x + 3(\sin x)$$

Practise 2.4

- Differentiate the following function

a) $y = (2x^2 - x)^2$

b) $y = \frac{1}{x^5} - \frac{2}{x^3} + \frac{x^3}{2}$

c) $y = 3\cos x - \sin x$

Differentiation Rules

3. Differentiation of Products

- Let u and v are differentiable function and if $y = uv$,

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Or

$$y' = uv' + vu'$$

Example

a) $y = x^3 \cos x$

$$\frac{dy}{dx} = x^3 \frac{d}{dx}[\cos x] + \cos x \frac{d}{dx}[x^3]$$

$$= -x^3 \sin \mathrm{x} + 3x^2 \cos \mathrm{x}$$

a) $y = 5x \sin x$

$$\frac{dy}{dx} = 5x \frac{d}{dx}[sin\ x] + \sin x \frac{d}{dx}[5x]$$

$$= 5x\ cosx + 5 \sin x$$

c) $y = x^2 \ln x$

$$\frac{dy}{dx} = x^2 \frac{d}{dx}[\ln x] + \ln x \frac{d}{dx}[x^2]$$

$$= x^2 \left(\frac{1}{x}\right) + \ln \mathrm{x}\ (2\mathrm{x})$$

$$= x + 2x\ lnx$$

Differentiation Rules

4. Differentiation of Quotients

- Let u and $v \neq 0$ are differentiable function and if $y = \frac{u}{v}$,

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Or

$$y' = \frac{vu' - uv'}{v^2}$$

Example

a) $y = \frac{1}{1 - sinx}$

$$\frac{dy}{dx} = \frac{(1 - \sin x)\frac{d}{dx}[1] - 1\frac{d}{dx}[1 - sinx]}{(1 - \sin x)^2}$$

$$= \frac{\big(1 - \sin x\big)\big(0\big) + 1(\cos x)}{(1 - \sin x)^2} = \frac{\cos x}{(1 - \sin x)^2}$$

a) $y = \frac{2x+5}{3x+2}$

$$\frac{dy}{dx} = \frac{(3x + 2)\frac{d}{dx}[2x + 5] - 2x + 5\frac{d}{dx}[3x + 2]}{(3x + 2)^2}$$

$$= \frac{\big(3x + 2\big)\big(2\big) - (2x + 5)(3)}{(3x + 2)^2} = -\frac{11}{(3x + 2)^2}$$

Example

Given that $f(1) = -2, f'(1) = 3, g(1) = 5$ and $g'(1) = -4$. Find the value of $\frac{dy}{dx}$ when x = 1 for each of the following.

(a) $y = (x-2)f(x)$

Solution:

$$y = (x-2)f(x)$$

$$\frac{dy}{dx} = (x-2)f'(x) + f(x)(1)$$

$$\left.\frac{dy}{dx}\right|_{x=1} = (1-2)f'(1) + f(1)(1)$$

$$= -1(3) + (-2)$$

$$= -5$$

(b) $y = \frac{f(x)}{g(x)}$

- Solution:

$$y = \frac{f(x)}{g(x)}$$

$$\frac{dy}{dx} = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$$

$$\left.\frac{dy}{dx}\right|_{x=1} = \frac{g(1)f'(1) - f(1)g'(1)}{(g(1))^2}$$

$$= \frac{(5)(3) - (-2)(-4)}{(5)^2} = \frac{7}{25}$$

$y = \frac{x-2}{f(x)}$

Practise 2.5

1.Differentiate the following function

a) $y = (2 - x^3)(x^3 + x - 1)$

b) $y = (5x^5 + x^2)^2$

c) $y = \frac{2}{\sqrt{x}+3}$

d) $y = \frac{x^3+1}{3x^2}$

2. Given that $f(2) = 1, f'(2) = -4, g(2) = 5$ and $g'(2) = 2$. Find the value of $h'(2)$ for each of the following.

(a)$h(x) = \sqrt{f(x) + g(x)}$

(c) $h(x) = [f(x)g(x)]^5$

(b)$h(x) = g(2f(x))$

(d) $h(x) = \left[\frac{f(x)}{g(x)}\right]^5$

Answer :

$2a) -\frac{1}{\sqrt{6}}$

$c) -56250$

$b) -16$

$d) -\frac{22}{3125}$

Higher Order Differentiation

- If y = f(x) , then the sequence of the differentiation can be written as

$$
\begin{array}{llll}
y' & f(x)' & \dfrac{dy}{dx} & \dfrac{d}{dx}[f(x)] \\
y'' & f(x)'' & \dfrac{d^2y}{dx^2} & \dfrac{d^2}{dx^2}[f(x)] \\
y''' & f(x)''' & \dfrac{d^3y}{dx^3} & \dfrac{d^3}{dx^3}[f(x)] \\
y^{(4)} & f(x)^{(4)} & \dfrac{d^4y}{dx^4} & \dfrac{d^4}{dx^4}[f(x)] \\
y^{(n)} & f(x)^{(n)} & \dfrac{d^ny}{dx^n} & \dfrac{d^n}{dx^n}[f(x)]
\end{array}
$$

Example

- a) If $f(x) = 4x^4 - 2x^3 + 3$, find its first five derivatives.

$$f'(x) = 16x^3 - 6x^2$$
$$f''(x) = 48x^2 - 12x$$
$$f'''(x) = 96x - 12$$
$$f^{(4)}(x) = 96$$
$$f^{(5)}(x) = 0$$
$$\therefore f^{(n)}(x) = 0, iff\ n \geq 5$$

b) If $y = \frac{x^2}{x-2}$, find $\frac{dy}{dx}, \frac{d^2y}{dx^2}$.

$$\frac{dy}{dx} = \frac{(x-2)\frac{d}{dx}(x^2) - (x^2)\frac{d}{dx}(x-2)}{(x-2)^2} = \frac{x^2 - 4x}{(x-2)^2}$$

$$\frac{d^2y}{dx^2} = \frac{(x-2)^2\frac{d}{dx}(x^2 - 4x) - (x^2-4x)\frac{d}{dx}(x-2)^2}{(x-2)^4} = \frac{8}{(x-2)^3} = \left(\frac{2}{(x-2)}\right)^3$$

Example

- If $y = Ax + Bx^2$ where A and B are constants, show that

$$x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 0$$

Solution:

$$y = Ax + Bx^2$$

$$\frac{dy}{dx} = A + 2Bx$$

$$\frac{d^2y}{dx^2} = B$$

$$x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 2Bx^2 - 2x\left(A + 2Bx\right) + 2\left(Ax + Bx^2\right)$$

$$= 2Bx^2 - 2Ax - 4Bx^2 + 2Ax + 2Bx^2$$

$$= 0$$

Exercise

- Find y' and y''

a) $y = (x^3 + 5)(3x + 2)$

b) $y = \frac{1}{(x+1)^2(x+2)}$

- Find

a) $y'''(1)$, if $y = x^5 - \frac{1}{x^5}$ ans:270

b) $\left.\frac{d^5y}{dx^5}\right|_{x=2}$ if $y = \frac{6}{x^3}$ ans:$\frac{-15120}{256}$

- If $y = 3x^2 - 5x$ show that

$$y\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y + 5 = 6x$$

The Derivative of Exponential Function

Theorem 2.3
if $y = e^u$ and $\mathrm{u} = f(x)$, then by using Chain rule,

$$\frac{dy}{dx}[e^u] = e^u \frac{du}{dx}$$

Example:

a) $y = 3e^{3x}$

$u = 3x, \frac{du}{dx} = 3 \qquad \frac{dy}{du} = 3e^u$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 3e^u \cdot 3 = 9e^{3x}.$$

b) $y = e^{x^2+1}$

$$\frac{dy}{dx} = (2x)\ e^{x^2+1} = 2xe^{x^2+1}$$

c) $y = e^{5x+\cos x}$

$$\frac{dy}{dx} = (5 - sinx\ )e^{5x+\cos x}$$

d) $y = lne^{\sin x}$

$= sinx \ln e$

$= \sin x$

$$\frac{d}{dx}(sinx) = \cos x$$

1. $loga^b = b \log a$
2. $\ln e = log_e e = 1$

Example

e) $$y = \frac{e^{3x}}{\left(e^{x} + e^{-x}\right)}$$

Let $$u = e^{3x}$$

$$\frac{du}{dx} = 3e^{3x}$$

$$v = e^{x} + e^{-x}$$

$$\frac{dv}{dx} = e^{x} - e^{-x}$$

$$\frac{dy}{dx} = \frac{\left(e^{x} + e^{-x}\right)\left(3e^{3x}\right) - e^{3x}\left(e^{x} - e^{-x}\right)}{\left(e^{x} + e^{-x}\right)^{2}}$$

$$= \frac{3e^{4x} + 3e^{2x} - e^{4x} + e^{2x}}{\left(e^{x} + e^{-x}\right)^{2}}$$

$$= \frac{2e^{4x} + 4e^{2x}}{\left(e^{x} + e^{-x}\right)^{2}}$$

The Derivative of Exponential Function $f(x) = a^x$

Theorem 2.4
if u $= f(x)$, and $y = a^u$ then

$$\frac{dy}{dx}[a^u] = a^u \ln a \frac{du}{dx}$$

Example:

a) $y = 10^x$

$$\frac{dy}{dx} = 10^x \ln 10$$

b) $y = 10^{(2x-5)}$

Let $u = 2x - 5,\ y = 10^u$

$$\frac{dy}{dx} = 10^{(2x-5)} \ln 10(2)$$
$$= 2(10^{2x-5}) \ln 10$$

c) $y = 10^{(2x-5)^{20}}$

Let $u = 2x - 5, v = u^{20}$ and $y = 10^v$

$$\frac{du}{dx} = 2, \qquad \frac{dv}{du} = 20u^{19}, \qquad \frac{dy}{dv} = 10^v ln10$$

$$\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{du} \cdot \frac{du}{dx}$$

$$= (10^v ln10) \cdot 20u^{19} \cdot 2$$
$$= (10^{(2x-5)^{20}} ln10). 20(2x-5)^{19} \cdot 2$$
$$= (10^{(2x-5)^{20}} ln10). 40(2x-5)^{19}$$

Practise 2.6

- a) $y = 2e^x + \ln x$
- b) $y = xln\ 3 + 3e^x$
- c) $y = e^{x^3+x}$
- d) $y = 3e^{2x} + 4x$
- e) $y = \sin x\ - e^{\sin x}$
- f) $y = 10^{(4x-5)^{10}}$
- g) $y = 2^{5x} \cdot 3^{4x^2}$

Answer

a) $2e^x + \frac{1}{x}$

b) $\ln 3 + 3e^x$

c)$(3x^2 + 1)e^{x^3+x}$

d) $6e^{2x} + 4$

e) $\cos x - \cos x e^{sinx}$

f) $40(4x-5)^9 10^{(4x-5)^{10}} ln10$

g) $2^{5x} \cdot 3^{4x^2}(8xln3 + 5ln2)$

The Derivative of Logarithm Function

Theorem 2.4
if $y = \ln x$ and $u = f(x)$, then

$$\frac{dy}{dx}[\ln u] = \frac{1}{u}\frac{du}{dx}$$

Example:

a) $y = ln3x$ $\quad \frac{dy}{dx} = 3 \cdot \frac{1}{3x} = \frac{1}{x}$

b) $y = x \ln 4 + 4 \ln x$ $\quad \frac{dy}{dx} = \ln 4 + \frac{4}{x}$

c) $y = 3\ln(x^2 + 2x)$ $\quad \frac{dy}{dx} = (3)(2x + 2) \cdot \frac{1}{x^2+2x}$

$$= \frac{6(x+1)}{x(x+2)}$$

d) $y = lne^x + \ln x^2$ $\quad \frac{dy}{dx} = 1 + \frac{2}{x}$

$= x \ln e + 2\, lnx$

$= x + 2 \ln x$

Example

e) $y = ln\left(\frac{3x+1}{x+1}\right)$ can be written as $ln(3x+1) - ln(x+1)$

$$\frac{dy}{dx} = \frac{1}{3x+1}(3) \cdot - \frac{1}{x+1}(1)$$

$$= \frac{3}{3x+1} - \frac{1}{x+1}$$

f) $y = ln\sqrt{\frac{1+2x}{1-2x}}$

Practise 2.7

a) $y = \ln(\ln x) + xln\ 2$

b) $y = \ln(x + 1) + \frac{1}{x+1}$

c) $y = \ln(e + x) + e \ln x$

d) $y = 3 \ln x + \ln 3x + xln\ 3$

e) $y = \ln e^{x^2}$

Answer

a) $\frac{1}{x \ln x} + \ln 2$

b) $\frac{x}{(x+1)^2}$

c) $\frac{1}{e+x} + \frac{e}{x}$

d) $\frac{3}{x} + \frac{1}{x} + \ln 3$

e) $2x$

The Derivative of Logarithm Function

Theorem 2.4

if $u = f(x)$, then

$$\frac{dy}{dx}[log_a u] = \frac{1}{u} log_a e \frac{du}{dx}$$

Example:

$a)\ y = log_2 x^5$

$u = x^5$

$\frac{du}{dx} = 5x^4$

$$\frac{dy}{dx} = \frac{d}{dx} log_2 x^5$$

$$= \frac{1}{x^5}(log_2 e)(5x^4)$$

$$= \frac{1}{x^5}\left(\frac{log_e e}{log_e 2}\right)(5x^4)$$

$$= \frac{1}{x^5}\left(\frac{lne}{\ln 2}\right)(5x^4) = \frac{5}{x \ln 2}$$

Change base of log

$$\log_b(a) = \frac{\log_x(a)}{\log_x(b)}$$

Example

b) $y = log\left(\sqrt{\frac{x^2+1}{1-x^2}}\right)$ can be written as $\frac{1}{2} log\left(\frac{x^2+1}{1-x^2}\right)$

$$\frac{1}{2} log\left(\frac{x^2+1}{1-x^2}\right) = \frac{1}{2}[log(x^2+1) - log(1-x^2)]$$

$$= \frac{1}{2}\left[\left(\frac{1}{(x^2+1)}\right)\log e(2x) - \left(\frac{1}{1-x^2}\right) loge(-2x)\right]$$

$$= \frac{1}{2}\left[\left(\frac{1}{(x^2+1)}\right)\frac{log_e e}{log_e 10}(2x) - \left(\frac{1}{1-x^2}\right)\frac{log_e e}{log_e 10}(-2x)\right]$$

$$= \frac{1}{2}\left[\left(\frac{1}{(x^2+1)}\right)\frac{2x}{ln10} - \left(\frac{1}{1-x^2}\right)\frac{-2x}{ln10}\right] = \frac{2x}{1-x^4}\log e$$

Example

b) $y = log\left(\sqrt{\frac{x^2+1}{1-x^2}}\right)$ can be written as $\frac{1}{2}log\left(\frac{x^2+1}{1-x^2}\right) =$

$$u = \frac{x^2+1}{1-x^2}$$

$$\frac{du}{dx} = \frac{4x}{(1-x^2)^2}$$

$$\frac{dy}{dx} = \frac{1}{2}\frac{d}{dx}log\frac{x^2+1}{1-x^2}$$

$$= \left(\frac{1}{2}\right)\left(\frac{1}{\frac{x^2+1}{1-x^2}}\right)\log e\left(\frac{4x}{(1-x^2)^2}\right)$$

$$= \left(\frac{1}{2}\right)\left(\frac{1-x^2}{1+x^2}\right)\log e\left(\frac{4x}{(1-x^2)^2}\right)$$

$$= \frac{2x}{1-x^4}\log e$$

Practise 2.8

Find the derivative

a) $y = log_3(4 - 9x)$
b) $y = \log(x^2 - 1)$
c) $y = log(x^3 + 4x)$
d) $y = \log\frac{x^2}{\sqrt{4-x^2}}$

Practise 2.8

Find the derivative

a) $y = log_3(4 - 9x)$

b) $y = \log(x^2 - 1)$

c) $y = log(x^3 + 4x)$

d) $y = \log\frac{x^2}{\sqrt{4-x^2}}$

$$\log\frac{x^2}{\sqrt{4-x^2}} = \log x^2 - \log\sqrt{4-x^2}$$

$$\frac{dy}{dx} = \frac{d}{dx}\log x^2 - \frac{1}{2}\frac{d}{dx}\log(4-x^2)$$

$$\frac{dy}{dx} = \frac{2\cancel{x}}{\cancel{x^2}^{\,1}}\log e - \left[\frac{1}{\cancel{2}}\left(\frac{-\cancel{2}x}{4-x^2}\right)\log e\right]$$

$$= \frac{2}{x}\log e + \frac{x}{4-x^2}\log e$$

$$= \log e\left(\frac{2}{x} + \frac{x}{4-x^2}\right)$$

$$= \log e\left(\frac{2(4-x^2)+x^2}{x(4-x^2)}\right)$$

$$= \frac{8-x^2}{x(4-x^2)}\log e \#$$

Answer

a) $\dfrac{-9}{(4-9x)ln\ 3}$

b) $\dfrac{2x}{(x^2-1)\ \ln 10}$

c) $\dfrac{3x^2+4}{(x^3+4x)\ \ln 10}$

d) $\dfrac{8-x^2}{x(4-x^2)}\log e$

Differentiation Of Trigonometric Functions

Original Rule	Generalized Rule (Chain Rule)
$\frac{d}{dx}\sin x = \cos x$	$\frac{d}{dx}\sin \mathbf{u} = \cos \mathbf{u}\,\frac{du}{dx}$
$\frac{d}{dx}\cos x = -\sin x$	$\frac{d}{dx}\cos \mathbf{u} = -\sin \mathbf{u}\,\frac{du}{dx}$
$\frac{d}{dx}\tan x = \sec^2 x$	$\frac{d}{dx}\tan \mathbf{u} = \sec^2 \mathbf{u}\,\frac{du}{dx}$
$\frac{d}{dx}\cot x = -\operatorname{cosec}^2 x$	$\frac{d}{dx}\cot \mathbf{u} = -\operatorname{cosec}^2 \mathbf{u}\,\frac{du}{dx}$
$\frac{d}{dx}\sec x = \sec x \tan x$	$\frac{d}{dx}\sec \mathbf{u} = \sec \mathbf{u} \tan \mathbf{u}\,\frac{du}{dx}$
$\frac{d}{dx}\operatorname{cosec} x = -\operatorname{cosec} x \cot$	$\frac{d}{dx}\operatorname{cosec} \mathbf{u} = -\operatorname{cosec} \mathbf{u} \cot \mathbf{u}\,\frac{du}{dx}$

Example

a) $y = 5x + \sin x$

$$\frac{dy}{dx} = \frac{d}{dx}[5x] + \frac{d}{dx}[\sin x] = 5 + \cos x$$

b) $y = \sin(2x - 5)$

Let $u = 2x - 5$, then $y = \sin u$

$$\frac{du}{dx} = 2, \qquad \frac{dy}{du} = \cos u$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$
$$= 2\cos u = 2\cos(2x - 5)$$

c) $y = sin^3\sqrt{2x - 5}$

Let $u = \sqrt{2x - 5}$, then $y = sin^3 u$

$$\frac{du}{dx} = \frac{1}{\sqrt{2x-5}}, \qquad \frac{dy}{du} = 3\ sin^2(u)\cos u$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$
$$= 3\ sin^2(u)\cos u \cdot \frac{1}{\sqrt{2x-5}}$$
$$= \frac{3\ sin^2\left(\sqrt{2x - 5}\right)\cos\left(\sqrt{2x - 5}\right)}{\sqrt{2x - 5}}$$

Let $w = \sin u\,, y = w^3$

$$\frac{dw}{du} = \cos u\,, \quad \frac{dy}{dw} = 3w^2$$

$$\frac{dy}{du} = \frac{dy}{dw} \cdot \frac{dw}{du}$$
$$= 3w^2 \cos u$$
$$= 3sin^2 u cos\ u$$

Example

d) $y = \cos(x^2 - 3)$

Let $u = x^2 - 3$, then $y = cos\, u$

$$\frac{du}{dx} = 2x, \qquad \frac{dy}{du} = -\sin u$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= -2x\, sin\, u = -2x\, sin(x^2 - 3)$$

e) $y = cos^2(x^2 - 3)$

Let $u = x^2 - 3$, $v = \cos u$ then $y = v^2$

$$\frac{du}{dx} = 2x, \qquad \frac{dv}{du} = -\sin u, \quad \frac{dy}{dv} = 2\mathrm{v}$$

$$\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{du} \cdot \frac{du}{dx}$$

$$= 2v\,(-\sin u\,)(2x) = (-2\cos u \sin u)\,(2x)$$

$$= -2x \sin 2u = -2x \sin 2(\,x^2 - 3)$$

$\sin(2a) = 2\sin(a)\cos(a)$

Example

f) $y = 4\tan\left(\frac{x}{2}\right)$

Let $u = \frac{x}{2}$, then $y = 4tan\, u$

$$\frac{du}{dx} = \frac{1}{2}, \qquad \frac{dy}{du} = 4\, sec^2 u$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= 4\, sec^2 u \cdot \frac{1}{2} = 2sec^2\left(\frac{x}{2}\right)$$

h) $y = \cot\ 5x \sec 7x$

By using product rules,

$$\frac{dy}{dx} = \frac{d}{dx}\cot 5x \sec 7x = \cot 5x \frac{d}{dx}\sec 7x + \sec 7x \frac{d}{dx}\cot 5x$$

$$= \cot 5x(7)\sec 7x tan\ 7x + \sec 7x(-5\ cosec^2 5x)$$
$$= \sec 7x(7\ cot5x \tan 7 - 5\ cosec^2 5x)$$

g) $y = x^2 \tan 2x$
By using product rules,

$uv' + vu'$
$u = x^2,\ v = \tan 2x$
$u' = 2x,\ v' = 2sec^2 2x$

$$\frac{dy}{dx} = \frac{d}{dx}x^2 \tan 2x = x^2 \frac{d}{dx}\tan 2\ x + \tan 2x \frac{d}{dx}x^2$$

$$= 2x^2 sec^2 2x + 2x \tan 2x$$

Example

i) $y = \cos^3(x^3 - 5x^2 + 2)$

Let $u = x^3 - 5x^2 + 2$ and $\mathrm{y} = cos^3 u$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Let $\mathrm{v} = \cos u,\ y = v^3$

$$\frac{dy}{du} = \frac{dy}{dv} \cdot \frac{dv}{du}$$

$$= 3v^2(-\sin u)$$

$$= 3(cos^2 u)(-\sin u)$$

$$= [3(\cos^2 u)(-\sin u)]\,(3x^2 - 10x) \cdot$$

$$= (-3)(3x^2 - 10x)[\cos^2(x^3 - 5x^2 + 2)\sin(x^3 - 5x^2 + 2)]$$

$$= (-9x^2 + 30x)[\cos^2(x^3 - 5x^2 + 2)\sin(x^3 - 5x^2 + 2)]$$

Practise 2.9

a) $cos^4 2x$

b) $sin^4\left(\frac{1}{2}x\right)$

c) $y = \sin x(1 + cosx)$

d) $y = \frac{1-\cos x}{\sin x}$

e) $y = 5xcosec\ 4x$

Answer

a) $-8\, cos^3 2x \sin 2x$

b) $2sin^3 \frac{1}{2}x \cos \frac{1}{2}x$

c) $\cos x(1+\cos x) - sin^2 x$

d) $\frac{1}{1+\cos x}$

e) $5(1-4x \cot 4x)\, cosec\, 4x$

Implicit Differentiation

The functions that we have met so far can be described by expressing one variable explicitly in terms of another variable—for example,

$$y = \sqrt{x^3 + 1} \quad \text{or} \quad y = x \sin x$$

or, in general, $y = f(x)$.

Some functions, however, are defined implicitly by a relation between x and y such as

1 $$x^2 + y^2 = 25$$

or

2 $$x^3 + y^3 = 6xy$$

For equation #2 ($x^3 + y^3 = 6xy$) t's not easy to solve for y explicitly as a function of x. Expressions obtained are very complicated.

Fortunately, we don't need to solve an equation for y in terms of x in order to find the derivative of y. Instead we can use the method of **implicit differentiation**.

This consists of differentiating both sides of the equation with respect to x and then solving the resulting equation for y'.

Example

(a) If $x^2 + y^2 = 25$, find $\frac{dy}{dx}$

Solution 1:

(a) Differentiate both sides of the equation $x^2 + y^2 = 25$:

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0$$

We can take the derivative of x^2 with respect to x but not y^2.
With y^2 we have to take the derivative with respect to y.

Continue..

Remembering that *y* is a function of *x* and using the Chain Rule, we have

$$\frac{d}{dx}(y^2) = \frac{d}{dy}(y^2)\,\frac{dy}{dx}$$

$$= 2y\,\frac{dy}{dx}$$

Thus

$$2x + 2y\,\frac{dy}{dx} = 0$$

Take the derivative and leave dy/dx in the answer

Now we solve this equation for *dy*/*dx*:

$$\frac{dy}{dx} = -\frac{x}{y}$$

Example

(b) $3y^2 - 2x^2 = 2xy$

We cannot write explicitly u in terms of x, Therefore, we differentiate it implicitly for both sides with respect to x

$$\frac{d}{dx}[3y^2 - 2x^2] = \frac{d}{dx}[2xy]$$

$$\frac{d}{dx}[3y^2] - \frac{d}{dx}[2x^2] = \frac{d}{dx}[2xy]$$

On the right side, we've got to recognize that we've actually got a product here, the 2x and the y(x) . So, to do the derivative of the right side we'll need to do the product rule

$$6y\frac{dy}{dx} - 4x = 2x\frac{dy}{dx} + 2y$$

Continue...

$$(6y - 2x)\frac{dy}{dx} = 2y + 4x$$

$$\frac{dy}{dx} = \frac{2y + 4x}{(6y - 2x)}$$

Simplify

$$\frac{dy}{dx} = \frac{y + 2x}{3y - x}$$

Example

c) $\dfrac{x}{y} - 3x^2 + ye^{x^2} = 1$

$$\frac{d}{dx}\left(\frac{x}{y}\right) - \frac{d}{dx}\left(3x^2\right) + \frac{d}{dx}\left(ye^{x^2}\right) = \frac{d(1)}{dx}$$

$$\frac{y\dfrac{d(x)}{dx} - x\dfrac{dy}{dx}}{y^2} - 6x + y\frac{d}{dx}\left(e^{x^2}\right) + e^{x^2}\frac{dy}{dx} = 0$$

$$y - x\frac{dy}{dx} - 6xy^2 + y^3\left(e^{x^2}.2x\right) + y^2e^{x^2}\frac{dy}{dx} = 0$$

$$\left(y^2e^{x^2} - x\right)\frac{dy}{dx} = 6xy^2 - y - 2xy^3e^{x^2}$$

$$\frac{dy}{dx} = \frac{6xy^2 - y - 2xy^3e^{x^2}}{y^2e^{x^2} - x}$$

$d)\ \ln(xy) - 3y^2 = 2x^3$

$$\ln x + \ln y - 3\left(y^2\right) = 2x^3$$

$$\frac{1}{x} + \frac{1}{y}\frac{dy}{dx} - 6y\frac{dy}{dx} = 6x^2$$

$$\left(\frac{1}{y} - 6y\right)\frac{dy}{dx} = 6x^2 - \frac{1}{x}$$

$$\frac{dy}{dx} = \frac{6x^2 - \dfrac{1}{x}}{\left(\dfrac{1}{y} - 6y\right)}$$

Example

e) $x^2y^3 + \tan\left(x^2y\right) = \ln x$

$$\frac{d}{dx}\left(x^2y^3\right) + \frac{d}{dx}\left[\tan\left(x^2y\right)\right] = \frac{dy}{dx}\ln x$$

$$x^2\frac{d}{dx}\left(y^3\right) + y^3\frac{d}{dx}\left(x^2\right) + \sec^2\left(x^2y\right)\frac{d}{dx}\left(x^2y\right) = \frac{1}{x}$$

$$x^2\cdot 3y^2\frac{dy}{dx} + y^3\cdot 2x + \sec^2\left(x^2y\right)\left[x^2\frac{dy}{dx} + y\left(2x\right)\right] = \frac{1}{x}$$

$$\left[3x^2y^2 + x^2\sec^2\left(x^2y\right)\right]\frac{dy}{dx} = \frac{1}{x} - 2xy\sec^2\left(x^2y\right) - 2xy^3$$

$$\frac{dy}{dx} = \frac{\frac{1}{x} - 2xy\sec^2\left(x^2y\right) - 2xy^3}{\left[3x^2y^2 + x^2\sec^2\left(x^2y\right)\right]}$$

Example

(f) If $y = \frac{A\cos 2x + B\sin 2x}{x}$, where A and B are constants, show that

$$x\frac{d^2y}{dx} + 2\frac{dy}{dx} + 4xy = 0$$

The given expression can be written as

$$xy = A\cos 2x + B\sin 2x$$

By using implicit differentiation we obtain

$$y + x\frac{dy}{dx} = -2A\sin 2x + 2B\cos 2x$$

Continue…

Differentiate the above expression with respect to x we obtain

$$\frac{dy}{dx} + \frac{dy}{dx} + x\frac{d^2y}{dx} = -4\mathrm{A}\cos 2\mathrm{x} - 4\mathrm{B}\sin 2\mathrm{x}$$

$$= -4(A\cos 2x + Bsin\ 2x)$$

$$= -4xy$$

Finally we obtain

$$x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4xy = 0$$

Example

(g) $y = x^x$

By taking natural logarithm for both sides ,

$$\ln y = \ln x^x = x \ln x$$

Differentiate both sides,

$$\frac{d}{dx}[\ln y] = \frac{d}{dx}[x \ln x]$$

$$\frac{1}{y}\frac{dy}{dx} = x\frac{1}{x} + lnx(1)$$

$$\frac{1}{y}\frac{dy}{dx} = 1 + lnx(1)$$

$$\frac{dy}{dx} = y(1 + lnx)$$

$$\frac{dy}{dx} = x^x(1 + lnx)$$

Natural logarithm rules and properties

Rule name	Rule
Product rule	$\ln(x \cdot y) = \ln(x) + \ln(y)$
Quotient rule	$\ln(x / y) = \ln(x) - \ln(y)$
Power rule	$\ln(x^y) = y \cdot \ln(x)$
ln derivative	$f(x) = \ln(x) \Rightarrow f'(x) = 1 / x$
ln integral	$\int \ln(x)dx = x \cdot (\ln(x) - 1) + C$

Example

(h) $y = a^{x^3}$

By taking natural logarithm for both sides ,

$$\ln y = \ln a^{x^3} = x^3 \ln a$$

Differentiate both sides,

$$\frac{d}{dx}[\ln y] = \frac{d}{dx}[x^3 \ln a]$$

$$\frac{1}{y}\frac{dy}{dx} = 3x^2 \ln a$$

$$\frac{dy}{dx} = 3x^2 y \ln a$$

$$\frac{dy}{dx} = 3x^2 a^{x^3} \ln a$$

Natural logarithm rules and properties

Rule name	Rule
Product rule	$\ln(x \cdot y) = \ln(x) + \ln(y)$
Quotient rule	$\ln(x / y) = \ln(x) - \ln(y)$
Power rule	$\ln(x^y) = y \cdot \ln(x)$
ln derivative	$f(x) = \ln(x) \Rightarrow f'(x) = 1 / x$
ln integral	$\int \ln(x)dx = x \cdot (\ln(x) - 1) + C$

Practice 2.10

Find $\frac{dy}{dx}$ for each of the following:

a) $3x^2 - xy + 3y = 7$

b) If $y = \frac{A\cos 3x + B\sin 3x}{x}$, where A and B are constants, show that

$$x\frac{d^2y}{dx} + 2\frac{dy}{dx} + 9xy = 0$$

c) $y = x^{sinx}$

d) $ye^{xy} = x^2 + 1$

e)$ln\left(\frac{x}{y}\right) - 3x^2 + e^{xy} = 1$

f) $sin(xy) + ysinx = x$

Answer

- a)$\frac{dy}{dx} = \frac{3x^2-18x+7}{(3-x)^2}$
- c)$x^{\sin x}\left(\frac{\sin x}{x} + \ln x \cos x\right)$
- d) $\frac{dy}{dx} = \frac{2x-y^2e^{2xy}}{e^{xy}(xy+1)}$
- e) $\frac{dy}{dx} = \frac{6x-\frac{1}{x}-ye^{xy}}{xe^{xy}-\frac{1}{y}}$
- f)$\frac{dy}{dx} = \frac{1-ycosx-ycos(xy)}{xcos(xy)+sinx}$

Differentiation of Parametric Function

If $x = h(t)$ and $y = g(t)$ then

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

Where $\dfrac{dt}{dx} = \dfrac{1}{\frac{dx}{dt}}$

Example

(a) Given $x = 2t$ and $y = 4 - 4t - 4t^2$

i) Find $\frac{dy}{dx}$ by using parametric differentiation

$$\frac{dy}{dt} = -4 - 8t, \quad \frac{dx}{dt} = 2$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$
$$= (-4 - 8t) \cdot \frac{1}{2}$$
$$= -2 - 4t$$

ii) y in terms of x. Hence, find $\frac{dy}{dx}$

From $x = 2t$, we obtain $t = \frac{x}{2}$,

Thus, $y = 4 - 2x - x^2$,

$$\therefore \frac{dy}{dx} = -2 - 2x$$

Example

- A curve has parametric equation

$$x = \frac{t-3}{t}, \qquad y = \frac{t^2+4}{t}$$

Evaluate the following at $t = 1$

a) $\frac{dy}{dx}$ b) $\frac{d^2y}{dx^2}$

SOLUTION:

a) $\frac{dy}{dt} = 1 - 4t^{-2}$, $\frac{dx}{dt} = 3t^{-2}$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= \left(1 - \frac{4}{t^2}\right) \cdot \frac{t^2}{3}$$

$$= \frac{1}{3}(t^2 - 4)$$

$$\frac{dt}{dx} = \frac{1}{\frac{dx}{dt}}$$

$If\ t = 1$

$$\frac{dy}{dx} = \frac{1}{3}(t^2 - 4)$$

$$= \frac{1}{3}(1^2 - 4)$$

$$= -1$$

Continue..

(b) Let $w = \frac{dy}{dx} = \frac{1}{3}(t^2 - 4)$ and $\frac{dw}{dt} = \frac{2}{3}t$, Hence,

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{dw}{dx} = \frac{dw}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dw}{dt} \cdot \frac{dt}{dx} = \left(\frac{2}{3}t\right) \cdot \left(\frac{t^2}{3}\right) = \frac{2}{9}t^3$$

When $t = 1$

$$\frac{d^2y}{dx^2} = \frac{2}{9}(1)^3 = \frac{2}{9}$$

Example

- Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for

$$x = 3^t, \qquad y = 9^t$$

Evaluate the following at $t = 1$

a) $\frac{dy}{dx}$ b) $\frac{d^2y}{dx^2}$

SOLUTION:

a) $y = 9^t = 3^{2t},$ $\frac{dx}{dt} = 3^t \ln 3$

$$\frac{dy}{dt} = 3^{2t}2ln3\,,$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dt}{dx} = \frac{1}{\frac{dx}{dt}}$$

$$= (3^{2t}2ln3) \cdot \frac{1}{3^t \ln 3}$$

$$= 2.3^t$$

$$If\ t = 1$$

$$\frac{dy}{dx} = 2.3^t = 2.3^1 = 6$$

Continue..

(b) Let $w = \frac{dy}{dx} = 2.3^t$ and $\frac{dw}{dt} = 2.3^t ln3$, Hence,

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{dw}{dx} = \frac{dw}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dw}{dt} \cdot \frac{dt}{dx} = (2.3^t ln3) \cdot \left(\frac{1}{3^t \ln 3}\right) = 2$$

Practice 2.11

a) Find $\frac{dy}{dx}$ in terms of t for the curve with parametric equations

$$x = 3t^2 - 3t^3, \qquad y = t(t^2 + 3)$$

b) Find $\frac{dy}{dx}$ in terms of θ for a curve with parametric equations

$$x = \cos\theta\,, \qquad y = \sin\theta$$

Hence evaluate $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $\theta = \frac{\pi}{4}$.

c) Find $\frac{dy}{dx}$ for following parametric equation

$$x = 2t - lnt, \qquad y = t^2 - lnt^2$$

Ans:

a) $\frac{t^2+1}{t(2-3t)}$

b) $\frac{dy}{dx} = -cot\theta = -1$

$\frac{d^2y}{dx^2} = -cosec^3\theta = -2\sqrt{2}$

c) $\frac{2(t^2-1)}{2t-1}$


thank you!